Watch out Snapdragon!

Intel's latest Lakefield processors with Intel Hybrid Technology are all about enabling new compact form-factors with dual displays and low power consumption.

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We sit down with Intel's Gokul Subramaniam to get the low-down on Intel's Lakefield processors and to gain insight into the new device segments that Intel and Qualcomm are both competing for.

Gokul Subramaniam,
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So how's Lakefield different from all the older Intel processors that were built for compact form factors?

Gokul: What we launched is the Intel Core processors with Intel Hybrid Technology. It was codenamed Lakefield. The biggest thing is that it leverages Intel's Foveros 3D packaging technology which allows us to stack up and get a smaller footprint on the silicon. And it has the hybrid CPU architecture so we're able to achieve low power and performance scalability. More importantly, it opens up new avenues for devices in the sub-10-inch display segment. It also enables dual-display and foldable notebook form factors. We are also able to give core performance with full compatibility which is really critical for Windows, both in 32 and 64-bit versions.

Lakefield uses Sunny Cove and Tremont cores, and Intel stated that for the ISA, Lakefield will stick with Tremont. Since Sunny Cover is a newer architecture with more advanced features, are you withholding or scaling back by sticking to Tremont for the ISA?

Gokul: If you look at the segment itself, the architecture of having four Tremont and one Sunny Cove is primarily for us to drive the best of two architectures. With the hardware-guided scheduling and working with the Windows 10 real-time scheduler, it allows us to get a performance spread in the foreground that can be prioritised and driven by Sunny Cove so you get very high response and you can also leverage the Intel Turbo technology. Then, in the background and for low-priority tasks, you get to schedule with Tremont. The key balance here is to reduce the power consumption while completing the tasks in the background. So the current architecture of four Tremont and one Sunny Cove allows us to do the best for both worlds.

In terms of AI, we still use our GPU which is a Gen11 IGP. Gen11 can drive some of the AI workloads and AI-enhanced workloads such as video stylisation, analytics or image resolution upscaling. So we are able to get 2X throughput for AI-enhanced workloads using our Gen11 GPU. It's stuck to that because for this segment, the power and performance balance is important and so is the footprint of the silicon itself.

Intel had Amber Lake processors in the 7W TDP segment and Lakefield also goes into the same 7W segment. In Amber Lake, particularly the 8500Y, Intel had HyperThreading enabled whereas with Lakefield, none of the cores (Sunny Cove and Tremont) have HyperThreading enabled. Is this because of Foveros or will we see new SKUs with HyperThreading enabled later on?

Gokul: The key part is that for this segment i.e. the sub-10-inch devices which feature smaller form-factors and are optimised for low-power usage. The kind of combination of the IPs needed in the silicon is something we've picked by considering the user experience. We've picked the best of what needs to be used. There is no limitation with respect to Foveros. If you think about the device segment i.e. extremely ultra-mobile, thin laptops and companion devices which might be dual-displays and will probably use OLED panels, low-power consumption is very important and the kind of performance required for these scenarios is the reason why we chose these IPs.

Since Intel did not disclose performance numbers, we had to resort to leaks. We noticed on Geekbench the scores for the Lakefield processor seems to match the Snapdragon 835 and 855 in terms of multi-core and single-core performance respectively. Then there's Celadon, Intel's own open-source Android distribution. So if we were to don a conspiracy hat and put the two together, we'd say you're also gunning for the smartphone market.

Gokul: Let's talk about Lakefield compared to a Snapdragon 8CX. We have a hybrid architecture, with full Windows 10 compatibility and we've managed very small package sizes. Now when you look at how the Core i5 SKU of Lakefield compares to Snapdragon's 8CX, we're able to drive 3.8x faster multi-threaded performance, up to 39x better multitasking in productivity tasks and up to 20x faster web-browsing experience. And these allow us to put up very competitive, low-power, high-performance SKUs for that segment. As for the question about entering into the smartphone segment, I'll defer to our comms department.